Assignment 2

Topic: Speech Enhancement (Question 1).
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Github link: repository

1 Objective

The task is to enhance the speech of each speaker in a multi-speaker environment.

Download the VoxCeleb1 and VoxCeleb2 datasets using this link [5]. Take one pre-trained speaker verification model from 'HuBERT Large', 'wav2vec2 XLSR', 'UniSpeech SAT', and 'wavLM Base Plus' using this link and perform speaker verification (evaluation) using the list of trial pairs - VoxCeleb1 (cleaned) file. The dataset for audio files is available in the VoxCeleb1 folder.

Fine-tune the selected speaker verification model using LoRA (Low-Rank Adaptation) and ArcFace loss with the VoxCeleb2 dataset [6] available in the VoxCeleb2 folder. The first 100 identities (when sorted in ascending order) will be used for training, and the remaining 18 identities for testing. Compare the performance of the pre-trained and fine-tuned models on the list of trial pairs - VoxCeleb1 (cleaned) dataset using the following metrics: Equal Error Rate (EER in %)[7], True Accept Rate at 1% False Accept Rate (TAR@1%FAR), and Speaker Identification Accuracy[1].

To create a multi-speaker scenario dataset, mix/overlap utterances from two different speakers in the VoxCeleb2 dataset by following these instructions: Use the first 50 identities (when sorted in ascending order) of the provided VoxCeleb2 dataset (VoxCeleb2 folder containing metadata text files and audio files in `.m4a` format) to create a multi-speaker training scenario, and use the next 50 identities to create a multi-speaker testing scenario. Refer to this GitHub repository for mixing speaker utterances.Utilize the pre-trained SepFormer model to perform speaker separation and speech enhancement on the created test set, analyzing the following metrics: Signal to Interference Ratio (SIR), Signal to Artifacts Ratio (SAR), Signal to Distortion Ratio (SDR), and Perceptual Evaluation of Speech Quality (PESQ).Next, use the pre-trained and fine-tuned speaker identification models obtained in step II to determine which enhanced speech corresponds to which speaker after speaker separation. Report the Rank-1 identification accuracy for both models.

Finally, design a novel pipeline or algorithmic approach to integrate the speaker identification model with the SepFormer model to perform speaker separation and speech enhancement simultaneously. Fine-tune or train this new pipeline on the training set of the created multi-speaker scenario dataset and evaluate it on the test set. Report the results using the same metrics as above (SIR, SAR, SDR, PESQ) and Rank-1 identification accuracy using both pre-trained and fine-tuned models.The report should include observations and a detailed analysis of the changes in performance throughout the experiment.

2 Question 1.1

The dataset was downloaded. After that, we obtained a text file [5] consisting of 36,711 pairs of data samples from the VoxCeleb1 dataset. These pairs were annotated with 0 and 1, where 0 indicates that they belong to different speakers, and 1 indicates that they belong to the same speaker.

3 Question 1.2

We used the WavLM Base Plus model[8], which we imported from the `transformers` library along with its checkpoints. This provided us with a pre-trained WavLM model. We initially performed speaker verification using the list of trial pairs from the text file and obtained the results. Next, we fine-tuned the model using LoRA (Low-Rank Adaptation)[3] and ArcFace loss [4].

LoRA is a technique designed to efficiently fine-tune large-scale models by reducing the number of trainable parameters. Instead of modifying the entire weight matrix of a neural network layer, LoRA introduces low-rank matrices that approximate weight updates. This significantly reduces computational and memory overhead while maintaining performance. By freezing the original model weights and learning only the low-rank updates, LoRA enables effective adaptation of the model without excessive parameter growth.

In the fine-tuning process, we incorporated ArcFace loss, which is an improved version of softmax loss that enhances the discriminative power of deep learning models for face and speaker verification tasks. ArcFace loss works by introducing an additive angular margin penalty to the feature embeddings before the final classification layer. This increases inter-class variance while ensuring intra-class compactness, making speaker embeddings more separable. In our case, ArcFace was applied before the final sigmoid activation, ensuring improved speaker verification performance.

After applying LoRA, we significantly reduced the number of trainable parameters, as shown in Table 1.

Table 1: Comparison of Parameters Before and After Applying LoRA

Model	**Total Parameters**	**Trainable Parameters**
Original Model	94,381,936	94,381,936
Fine-tuned with LoRA	97,113,044	2,731,108

Using LoRA and ArcFace loss, we trained the model. However, after a certain number of iterations, the loss stopped decreasing and appeared to saturate. Since the batch size was set to 1, training with a single epoch was a reasonable trade-off. Finally, the fine-tuned model was evaluated on the list of trial pairs, and the performance metrics for both the pre-trained and fine-tuned models are presented in Table 2.

Table 2: Performance Comparison of Pre-trained and Fine-tuned Models

Model	**EER (%)**	**TAR@1% FAR**	**Speaker Identification Accuracy (%)**
Pre-trained	36.70	7.27	63.29
Fine-tuned	18.80	18.14	81.20

The results clearly indicate that fine-tuning was effective. By adjusting the number of trainable parameters in the LoRA framework, further improvements could be achieved due to its modular and adaptable design.

4 Question 1.3

We created a multi-speaker scenario by overlapping utterances from two randomly selected speakers. Since the audio samples had different lengths, the longer audio was truncated to match the length of the shorter one. This ensured that the original and mixed audio samples had the same duration. A total of 500 mixed audio files were generated. We then used SepFormer[9] from Hugging Face to separate the combined audio clips.

SepFormer is a deep learning model designed for speech separation, employing a dual-path transformer-based architecture to effectively isolate multiple speakers' voices from a mixed audio signal. Unlike traditional separation models that rely on recurrent or convolutional layers, SepFormer utilizes self-attention mechanisms in both intra-segment and inter-segment processing, allowing it to capture long-range dependencies and achieve superior separation performance. This makes it highly effective in disentangling overlapping speech signals.

Once the audio files were separated, we evaluated their quality using the following metrics:

1. **Signal to Interference Ratio (SIR):** Measures how well the target signal is preserved while suppressing interfering sources. Higher values indicate better separation.

2. **Signal to Artifacts Ratio (SAR):** Assesses the level of processing artifacts introduced during separation. Higher values indicate fewer distortions.

3. **Signal to Distortion Ratio (SDR):** Quantifies the overall distortion in the separated signal, including both interference and artifacts. Higher values indicate better separation quality.

4. **Perceptual Evaluation of Speech Quality (PESQ):** A perceptual metric that evaluates speech quality as heard by humans, ranging from -0.5 (poor quality) to 4.5 (excellent quality).

The evaluation results are shown in Table 3.The results indicate that SepFormer is effective in separating speakers from mixed audio, as evidenced by the high SIR value. However, the presence of artifacts (SAR) and distortions (SDR) suggests that the model does not achieve perfect reconstruction. The moderate PESQ score further supports this observation, implying that while the separated speech is recognizable, it may not fully retain the natural characteristics of the original speakers. This performance aligns with the expected behavior of deep learning-based separation models, where there is often a trade-off between interference suppression and distortion introduction. Future improvements could involve fine-tuning SepFormer on domain-specific data, incorporating post-processing techniques for enhancement, or exploring alternative architectures to minimize artifacts and improve overall speech quality.

Table 3: Performance Metrics for Speech Separation

Metric	**Average Score**
SDR (dB)	13.00
SIR (dB)	22.61
SAR (dB)	14.25
PESQ	2.18

After obtaining the separated audio files, we classified them using both the pre-trained and fine-tuned models. The classification was performed by comparing the similarity of the generated audio files to the class audio files. The label assigned to each sample was based on the highest similarity score. The results are presented in Table 4.

Table 4: Rank-1 Accuracy for Speaker Classification

Model	**Rank-1 Accuracy**
Pre-trained Model	0.0876
Fine-tuned Model	0.2445

While fine-tuning improved accuracy, the performance gain was limited. This could be attributed to the fact that the model was not explicitly fine-tuned for classifying separated speech signals, especially when dealing with 50 test classes. Additionally, even though humans can recognize that the separated speech retains the original identity, many signal properties differ from the original audio, making classification challenging. Sample cases, including original, mixed, and separated audio files, have been uploaded to the GitHub repository linked at the beginning of this report.

5 Question 1.4

The proposed hybrid architecture aims to integrate the speaker identification model with the SepFormer model to perform both speaker separation and speech enhancement. The idea is to use the SepFormer model to first isolate individual speaker voices from a mixed audio signal and then apply the speaker identification model to classify each separated speech signal. This requires designing a novel pipeline that incorporates both models efficiently. One major challenge in implementing this approach is the high computational cost. SepFormer already operates on a transformer-based architecture, which requires significant memory and processing power. Adding a speaker identification model on top of this increases the computational burden further. Fine-tuning the entire pipeline would require extensive hyperparameter tuning, large-scale data augmentation, and empirical evaluation to ensure optimal performance.

Moreover, speaker separation introduces distortions and artifacts, which could negatively impact the identification model's accuracy.

Another critical aspect is the need for extensive empirical analysis. Since SepFormer is optimized for separation rather than speaker identity preservation, the fine-tuned speaker identification model may struggle to classify the separated speech correctly. To mitigate this, additional training strategies such as joint optimization of separation and classification losses, domain adaptation, or embedding alignment techniques may be required. However, these enhancements would further increase the complexity and computational demands of the system. Given these challenges, while a hybrid approach combining SepFormer and speaker identification models is theoretically feasible, its practical implementation would require substantial computational resources and fine-tuning efforts. Future work could explore efficient alternatives such as lightweight speech enhancement models, contrastive learning techniques, or knowledge distillation to reduce computational overhead while maintaining classification accuracy.

References

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